raffle.fun

EXPLANATION

How raffle.fun works

How an onchain raffle turns an unsold NFT into either a sale at your number or income while it waits.

READER

Anyone who knows NFTs but not Solidity

PROTOCOL COMMIT

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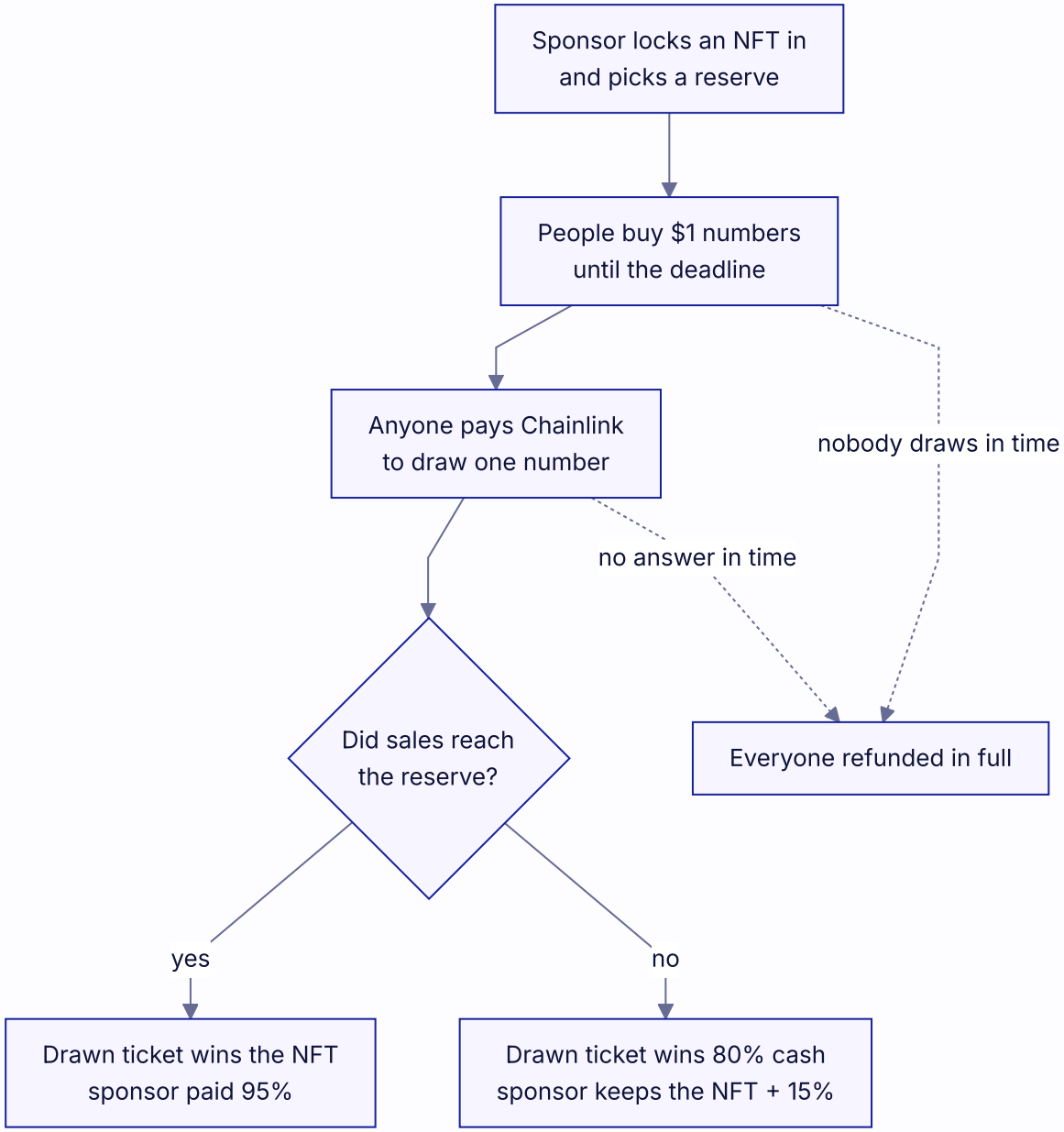
NOT DEPLOYED

NOT INDEPENDENTLY AUDITED

Pre-release

raffle.fun is a simple Ethereum mechanism for raffling one ERC-721 NFT with one-dollar USDC entries and a reserve price chosen in advance.

This article explains the candidate v1 design. The Solidity and canonical technical documents remain authoritative, and the protocol is not deployed or independently audited.



The whole mechanism end to end. Only two things are ever uncertain: how many entries sell, and which number gets drawn.

The sponsor's choice

The sponsor chooses:

- the NFT;
- the number of one-dollar entries required to meet the reserve;
- a sale end time, no more than 30 days away.

Those values become permanent when the raffle is created. The factory makes a cheap ERC-1167 clone, registers it, and escrows the NFT in one transaction. If the transfer or verification fails, the entire creation reverts.

The reserve is not a sell-out cap. A raffle with a 100,000-entry reserve keeps accepting entries until its deadline. Selling exactly 100,000 meets the reserve; selling more is also an NFT result.

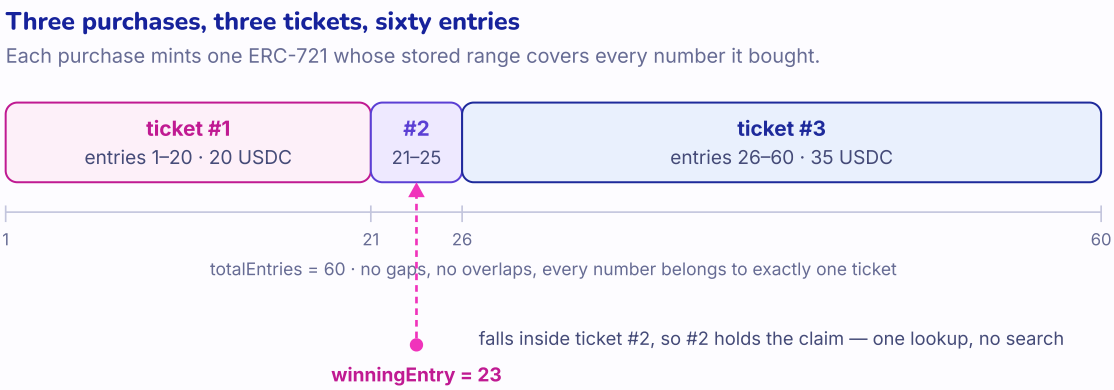
A ticket can contain many numbers

Every number costs exactly 1 USDC. A buyer can buy one number or many:

- buy 1 and receive one ERC-721 ticket for number 1;
- buy 20 and receive one ERC-721 ticket for a 20-number range;
- buy again later and receive another ticket for the next range.

Ticket IDs are simple counters: `1`, `2`, `3`, and so on. The contract stores each ticket's first and last entry separately. It does not create one NFT or storage record for every number, so buying a million entries is still one ticket mint and constant onchain work. The buyer must of course have and approve a million USDC.

A whole ticket can be transferred in any raffle state until it is claimed and burned; its range cannot be split. The current owner holds the bearer claim.



Numbers are handed out in contiguous blocks, so the contract stores one range per purchase instead of one record per number. The draw picks a number, and exactly one ticket contains it.

The draw

After the sale, anyone can pay the current Chainlink VRF v2.5 native fee in ETH to request the draw. The protocol does not take that fee from the USDC pot.

The raffle uses Chainlink's official native direct-funding wrapper through the exact-pinned `@chainlink/contracts@1.5.0` consumer base, interface, and client library. It does not deploy its own oracle or wrapper, and it does not require a raffle-managed VRF subscription.

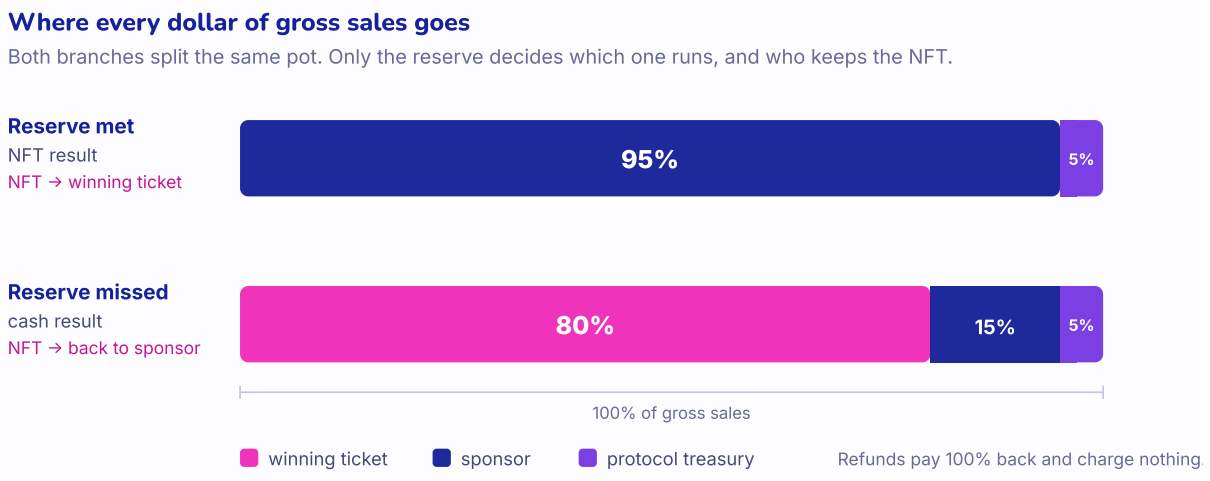
Chainlink later supplies one authenticated 256-bit word. The raffle maps it to:

```
winningEntry = (randomWord mod totalEntries) + 1
```

The callback records only that number and the economic result. It never searches all tickets or transfers assets. To claim, someone supplies a ticket ID; the contract loads its first and last number and checks whether the winner lies inside.

This is why entry count does not create a gas-limit problem:

OPERATION	WORK
buy any entry count	one payment and one ticket mint
draw callback	one modulo and bounded state writes
prove winner	load one ticket range
refund	loop only over submitted tickets, at most 100



The reserve decides the branch. Both branches pay the sponsor and the protocol; only the destination of the NFT and the drawn ticket's payout change.

If the reserve is met

The selected ticket wins the NFT. The USDC pot does not leave merely because the callback picked a winner. A later permissionless settlement proves the ticket and allocates the sponsor and protocol balances without reading its owner, burning it, or transferring an asset.

Settlement records:

- the protocol records a 5% balance for its immutable treasury;
- the protocol records a 95% balance for the sponsor's immutable recipient;
- the winning ticket ID is recorded while the ticket remains transferable.

Only the current ticket owner may redeem the NFT. `redeemWinningTicket` can perform settlement first when necessary, then burns the ticket and transfers the NFT in the same transaction. A failed transfer or ownership check reverts the burn. Sponsor proceeds and the protocol fee are separately releasable after settlement, so winner inactivity or a broken prize cannot block those quote claims.

This result is final. There is no deadline after which an unredeemed NFT turns back into refunds, so the winner's claim on the NFT stays open indefinitely — and so does the risk that the collection itself stops cooperating. If that collection is later paused, upgraded, restricted, or burned, delivery can fail forever while every USDC claim still settles normally. That is the single biggest thing to check before buying into a raffle: the NFT is only as reliable as the contract that issued it.

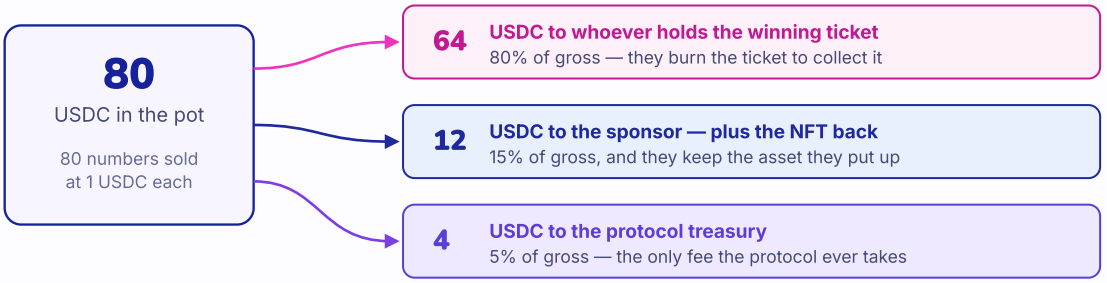
If the reserve is missed

Gross sales are split into fixed claims: the selected ticket carries the right to 80%, the protocol treasury receives 5%, and the sponsor receives the exact 15% remainder plus the NFT.

Cash settlement follows the same accounting rule: anyone may prove and record the winning ticket and allocate the 80/5/15 split without reading ownership. The ticket remains transferable until its current owner atomically burns it and receives the 80% cash amount. The sponsor and protocol claims remain independently releasable. The cash result is final; it does not later turn into refunds because winner inactivity affects no other participant.

A worked example: 80 entries sold, reserve was 100

The reserve was missed, so this is the cash result. Nothing is rounded away — the three shares add back to 80.

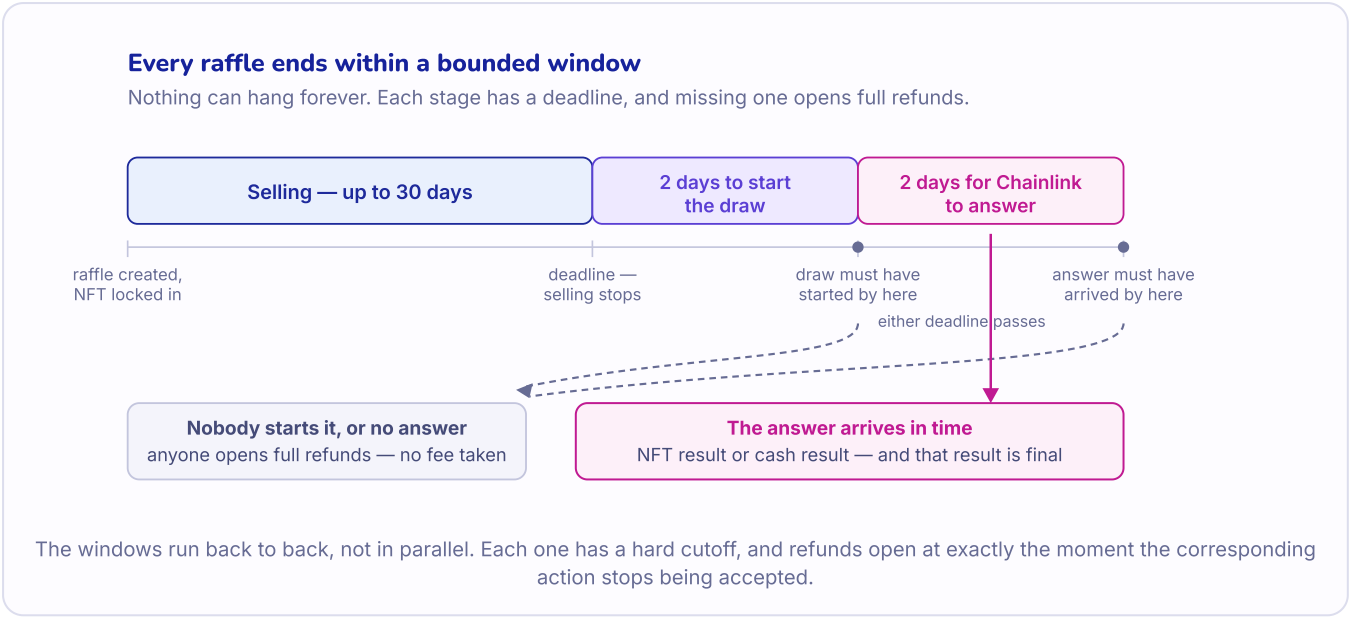


64 + 12 + 4 = 80 · had the reserve been met instead, the ticket would have won the NFT and the sponsor 76 USDC

Eighty one-dollar entries against a hundred-entry reserve. Each share is taken from gross sales, and the sponsor's share is the exact remainder, so nothing is lost to rounding.

If the process stalls

The draw request is available for two days after sale end. If nobody gets a request accepted by that hard deadline, anyone can open full, fee-free refunds. If a request is accepted, Chainlink gets a fresh two-day callback window; a callback received at or after that second deadline is ignored and refunds are available instead. A request in the final second can therefore make the maximum nominal path just under four days.



The windows run back to back, not in parallel. Each one has a hard cutoff, and refunds open at exactly the moment the corresponding action stops being accepted.

Each ticket refunds `number of entries × 1 USDC`. A 20-number ticket therefore refunds 20 USDC in one burn. The sponsor can reclaim the NFT independently.

A zero-sale raffle uses the same `Refunding` state with zero liability. Its sponsor may finalize it early, or anyone may finalize it after the deadline.

The callback boundary is deterministic: only a wrapper-authenticated callback before the deadline can resolve the raffle. At the deadline and afterward, an ABI-decodable callback is ignored even if `enableRefunds` has not yet been called.

What is and is not guaranteed

Every raffle clone is fixed and has no administrator. The factory is also ownerless, cannot be paused or reconfigured, and cannot alter existing prizes, reserves, owners, results, or claims. USDC movements use `SafeERC20` under the trusted canonical-token assumption, and known protocol addresses cannot receive tickets or payouts.

There is no administrator to abuse the system — and none to repair it. A bug found after launch cannot be patched into a running raffle, and new raffles cannot be halted onchain. The response available is to warn people, take the interface down, and ship a new factory; the raffles already running finish on their own terms.

The design still relies on external systems:

- the NFT contract can be upgradeable, frozen, dishonest, or broken, and after a result there is no refund to fall back on;
- Chainlink must provide correct, available VRF service, and somebody must be willing to pay the ETH fee that starts the draw — nobody is obliged to;
- Ethereum must continue including transactions; a request or callback that misses its cutoff cannot be replayed, and the raffle refunds instead;
- Circle can pause or blacklist USDC;
- users can lose keys, or send a ticket to a contract that can never act on it;

- no finite confirmation count guarantees unlimited economic value, and the contracts impose no cap on how large a raffle can get;
- chance-based promotions and gambling are legally regulated.

Those limits are why Sepolia testing, independent audit, operational monitoring, and jurisdiction-specific legal review remain release blockers.